



2011
TRIAL
HIGHER SCHOOL CERTIFICATE

GIRRAWEEEN HIGH SCHOOL

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board – approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

Attempt Questions 1 – 7
All questions are of equal value

Girraween High School Trial Higher School Certificate 2011

Total marks-84

Attempt all 7 questions

All questions are of equal value

Answer each question on a separate piece of paper clearly marked Question 1, Question 2, etc.

Each piece of paper should show your name.

Question 1 (12 Marks) Use a separate piece of paper		Marks
(a)	Factorise $x^3 + 64$	1
(b)	Find $\lim_{x \rightarrow 0} \frac{\tan 3x}{2x}$	2
(c)	Differentiate $y = x \tan^{-1} x$.	2
(d)	Solve the inequality $\frac{3}{x-1} \leq 2$	2
(e)	Find the point $P(x, y)$ that divides the interval A $(-2, 3)$ and B $(2, -5)$ externally in the ratio 2 : 1	2
(f)	Use the table of standard integrals to find $\int \sec 3x \tan 3x \, dx$	1
(g)	Use the substitution $u = x^3 - 1$ to evaluate $\int_0^2 \frac{x^2}{(x^3 - 1)^2} \, dx$	3

Question 2 (12 Marks) Use a separate piece of paper

- (a) The population P of Happytown has been falling at a rate proportional to the current population. That is P satisfies the equation $\frac{dP}{dt} = -kP$
- (i) Show that $P = P_0 e^{-kt}$ satisfies this equation 1
- (ii) In 1990 the population of Happytown was 25,000 by 2000 it had fallen to 20,000.
Find the values of P_0 and k . 2
- (iii) Find the population of Happytown in 2010. 2
- (iv) At what rate is the population falling in 2010. (to the nearest person) 1
- (b) The polynomial $P(x)$ has a remainder of -2 when divided by $(x - 2)$
and has a remainder of 4 when divided by $(x + 1)$. Find the remainder when
 $P(x)$ is divided by $(x - 2)(x + 1)$. 2
- (c) A skydiver jumps from an aircraft at 4000m above the ground. The velocity
 V m/s at which he falls is given by $V = 60(1 - e^{-0.16t})$
- (i) Find the acceleration 10 seconds after he jumps. (2 dec pl) 2
- (ii) Find the distance he has fallen in the first 10 seconds. (nearest metre) 2

Question 3 (12 Marks) Use a separate piece of paper.

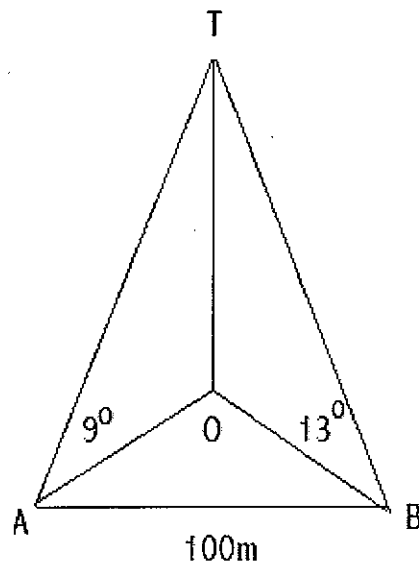
- (a) A chess club has 7 male and 5 female members. A grade team of 4 players is selected at random.

(i) In how many ways can the team be chosen. 1

(ii) What is the probability that the team contains a particular boy and particular girl. 2

(b) Using the substitution $t = \tan \frac{\theta}{2}$ or otherwise, show that $\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}$ 2

(c) From a point A due south of a tower OT the angle of elevation is 9°
100m from A at B that is due east of the tower, the angle of elevation is 13° .
As in the diagram below. Find the height of the tower. 3



(d) The polynomial $x^3 - 2x^2 - 5x + 7 = 0$ has roots α, β and γ

(i) Find the value of $\alpha + \beta + \gamma$ 1

(ii) Find the value of $\alpha\beta\gamma$ 1

(iii) Find the cubic equation with roots $\alpha + 2, \beta + 2, \gamma + 2$ 2

Question 4 (12 Marks) Use a separate piece of paper

- (a) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$
- Show that the equation of the tangent at P is $y = px - ap^2$ 1
 - Show that the point of intersection of the tangents from P and Q is $\{a(p+q), apq\}$ 2
 - The tangents from P and Q intersect at 45° . Show that $p - q = 1 + pq$ 1
- (b) The curve $y = \tan x + \sec x$ is rotated about the X axis between $x = \frac{\pi}{6}$ and $\frac{\pi}{3}$.
- Find the volume of the solid formed. 3
- (c) The 3rd and 5th term of a series are 10 and 4 respectively
- If the series is an A.P. find a the first term and d the common difference 2
 - Find the least number of terms of the A.P. for the sum to be negative. 1
 - If the series is a G.P. find a the first term and r the common ratio/s 2

Question 5 (12 Marks) Use a separate piece of paper

- (a) (i) Express $2\sin x - 3\cos x$ in the form $R\sin(x - \alpha)$ 2
- (ii) Hence or otherwise solve $2\sin x - 3\cos x = 2.5$, $0 \leq x \leq 360^\circ$ (nearest minute) 2
- (b) (i) Two fair six sided dice are rolled. Show that the probability of the total being 7 is $\frac{1}{6}$ 1
- (ii) If the dice are rolled seven times find the probability that the total is 7 exactly twice. 2
- (c) Find the greatest coefficient in the expansion of $(2x+3)^7$ 2
- (leave answer in unexpanded form)
- (d) (i) Use the double angle expansion to express $\cos 2\theta$ in terms of $\cos \theta$ 1
- (ii) Using part (i) or otherwise find the exact value of $\cos 36^\circ$ given $\cos 18^\circ = \frac{\sqrt{5}+1}{4}$ 2

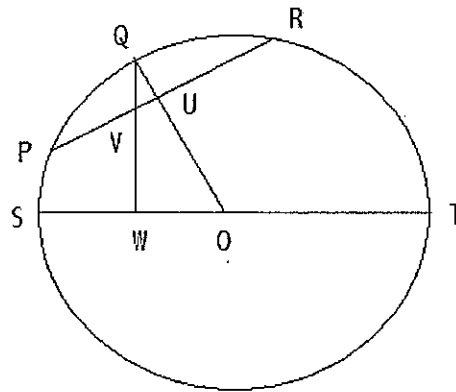
Marks

Question 6. (12 Marks) Use a separate piece of paper

(a) Use Mathematical Induction to show that for all integers $n \geq 1$ that

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \frac{4}{5!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!} \quad 3$$

(b) In the circle below Q is the mid point of the arc PR . The radius OQ meets the chord PR at U . From Q a perpendicular is dropped to W on the diameter ST meeting PR at V .



Prove that $OUVW$ is a cyclic quadrilateral.

3

(c) For the function defined by $f(x) = \frac{2x^2}{x^2 + 1}$

(i) Show that $f(x)$ is an even function 1

(ii) Show that $f(x)$ is monotonic increasing for $x \geq 0$ 2

(iii) Find the inverse function $f^{-1}(x)$ $x \geq 0$ 2

(iv) State the domain and range of $f^{-1}(x)$ 1

Question 7 (12 Marks) Use a separate piece of paper

(a) The velocity v m/s of a particle moving along the x -axis is given by $v^2 = 24 + 8x - 2x^2$

where x is the displacement from the origin.

- | | | |
|-------|---|---|
| (i) | Show that the motion is Simple Harmonic | 2 |
| (ii) | Find the centre of the motion | 1 |
| (iii) | Find the amplitude of the motion. | 1 |
| (iv) | Find the period of the motion. | 1 |
| (v) | Find the greatest velocity. | 1 |

(b) A cricketer at the centre of an oval strikes a cricket ball from the ground level producing a velocity of 30m/s. 60 meters from the batsman and in the same plane as the motion of the ball is a fence 1.2m high. The path of the ball is given by the parametric equations

$$x = vt \cos \theta \text{ and } y = vt \sin \theta - \frac{1}{2}gt^2 \text{ where } g = 10\text{ms}^{-2} \text{ (there is no need to prove this)}$$

(i) Show that the Cartesian equation of the path of the ball is given by

$$y = (\tan \theta)x - \frac{\sec^2 \theta}{180}x^2 \quad 2$$

(ii) Show that the angle of projection can be found by solving

$$20 \tan^2 \theta - 60 \tan \theta + 21.2 = 0 \quad 2$$

(iii) Find the range of angles for which the ball will clear the fence. 1

(iv) Find the maximum distance the ball can land beyond the fence. 1

END OF PAPER

H.S.C TRIAL EXTENSION 1

Question 1

a) $x^3 + 64 = (x+4)(x^2 - 4x + 16)$ ①

b) $\lim_{x \rightarrow 0} \frac{\tan 3x}{2x} = \lim_{x \rightarrow 0} \frac{\tan 3x}{3x} \cdot \frac{3}{2}$
 $= 1 \cdot \frac{3}{2}$
 $= \frac{3}{2}$ ①

c) $y = x \tan^{-1} x$
 let $u = x$ $v = \tan^{-1} x$
 $\frac{du}{dx} = 1$ $\frac{dv}{dx} = \frac{1}{1+x^2}$

$\frac{d(u \cdot v)}{dx} = v \cdot \frac{du}{dx} + u \cdot \frac{dv}{dx}$
 $= \tan^{-1} x + \frac{x}{1+x^2}$ ②

d) $\frac{3}{x-1} \leq 2$

$3(x-1) \leq 2(x-1)^2$
 $0 \leq 2(x-1)^2 - 3(x-1)$
 $0 \leq (x-1) \{ 2x-2-3 \}$
 $0 \leq (x-1)(2x-5)$

$x < 1$ $x > \frac{5}{2}$ ②

e) $A(-2, 3)$ $B(2, -5)$
 $2: -1$

$\left(\frac{2 \times 2 + (-2 \times -1)}{2-1}, \frac{-5 \times 2 + 3 \times -1}{2-1} \right)$
 $(6, -13)$ ②

(f) $\int \sec 3x \tan 3x dx$
 $= \frac{1}{3} \sec 3x + C$ ①

(g) $I = \int_0^2 \frac{x^2}{(x^3+1)^2} dx$

let $u = x^3 + 1$

$\frac{du}{dx} = 3x^2$

$du = 3x^2 dx$

when $x = 0$ $u = +1$
 $x = 2$ $u = 9$

$= \frac{1}{3} \int_0^2 \frac{3x^2 dx}{(x^3+1)^2}$

$= \frac{1}{3} \int_{+1}^9 \frac{du}{u^2}$ ③

$= \frac{1}{3} \left[-\frac{1}{u} \right]_{+1}^9$

$= \frac{1}{3} \left[-\frac{1}{9} + \frac{1}{+1} \right]$

$= +\frac{8}{27}$

Question 2.

(a) (i) $P = P_0 e^{-kt}$

$\frac{dP}{dt} = -k(P_0 e^{-kt})$
 $= -kP$ ①

(ii) $t = 0$ (1990) $P = 25,000$

$25000 = P_0 e^{-k(0)}$

$P_0 = 25000$ ①

$t = 10$ (2000) $P = 20,000$

$20000 = 25000 e^{-10k}$

$0.8 = e^{-10k}$

$\ln 0.8 = -10k \ln e$

$k = 0.022314$ ①

(iii) $t = 20$ (2010)

$P = 25000 e^{-20(0.022314)}$

$P = 16000$ ②

(iv) $\frac{dP}{dt} = -kP$ ①

$= -0.022314 \times 16000$

$= -357 \text{ people/year}$

(b) $P(x) = Q(x)(x-2)(x+1) + ax+b$

$P(2) = -2$

$-2 = Q(2)(0)(3) + 2a+b$

$-2 = 2a+b$ (α)

$P(-1) = 4$

$4 = Q(-1)(-3)(0) - a+b$

$4 = -a+b$ (β)

(α)-(β) $-6 = 3a$ $a = -2$

$\therefore b = 2$

$\therefore \text{Remainder} = -2x+2$ ②

(c) (i) $V = 60(1 - e^{-0.16t})$

$a = \frac{dV}{dt} = 60(0.16 e^{-0.16t})$

$a_{(t=10)} = 60(0.16 e^{-1.6})$
 $= 1.9382 \text{ m/s}$ ②
 $= 1.94 \text{ m/s}$ 2 D.P.

(ii) $x = \int v dt$

$x = 60 \int_0^{10} 1 - e^{-0.16t} dt$
 $= 60 \left[t + \frac{e^{-0.16t}}{0.16} \right]_0^{10}$

$= 60 \left[(10 + 1.26185) - (0 + 6.25) \right]$

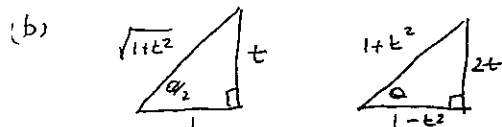
$= 300.711 \text{ m}$

$= 301 \text{ m}$ ②

Question 3.

(a) (i) ${}^{12}C_4 = 495 \text{ ways}$ (1)

(ii) $P(B+G+*+*) = \frac{{}^{10}C_2}{{}^{12}C_4}$
 $= \frac{45}{495}$
 $= \frac{1}{11}$ (2)



$\therefore \operatorname{cosec} \theta + \cot \theta = \frac{1+t^2}{2t} + \frac{1-t^2}{2t}$
 $= \frac{2}{2t}$
 $= \frac{1}{t}$
 $= \frac{1}{\tan \theta_2}$
 $= \cot \theta_2$ (2)

(c) $\tan \theta = \frac{h}{AO}$ $\tan 13 = \frac{h}{OB}$

$AO = \frac{h}{\tan \theta}$ $OB = \frac{h}{\tan 13}$

BY PYTHAGORAS IN $\triangle AOB$

$AO^2 + OB^2 = AB^2$
 $\frac{h^2}{\tan^2 \theta} + \frac{h^2}{\tan^2 13} = 10000$
 $h^2 \left(\frac{1}{\tan^2 \theta} + \frac{1}{\tan^2 13} \right) = 10000$
 $h^2 = \frac{10000}{58.6251614}$
 $h^2 = 170.57528$
 $h = 13.06 \text{ m.}$ (3)

(d) $x^3 - 2x^2 - 5x + 7 = 0$

α, β, γ

(i) $\alpha + \beta + \gamma = -\frac{b}{a} = 2$ (1)

(ii) $\alpha\beta\gamma = -\frac{d}{a} = -7$ (1)

(iii) WE REQUIRE $Y = X + 2$

$\therefore x = Y - 2$

$(Y-2)^3 - 2(Y-2)^2 - 5(Y-2) + 7 = 0$
 $Y^3 - 6Y^2 + 12Y - 8 - 2Y^2 + 8Y - 8 - 5Y + 10 + 7 = 0$

$Y^3 - 8Y^2 + 15Y + 1 = 0$ (2)

CHECK WHEN USING SUM OF ROOT $= -\frac{b}{a} = 8$.

WHICH IS EQUAL TO $\alpha + \beta + \gamma + 6$.

Question 4.

(a) (i) $x = 2ap$ $y = ap^2$

$\frac{dx}{dp} = 2a$ $\frac{dy}{dp} = 2ap$

$\frac{dy}{dx} = \frac{dy}{dp} \cdot \frac{dp}{dx} = p$

EQN

$Y - ap^2 = P(X - 2ap)$

$Y - ap^2 = PX - 2ap^2$

$Y = PX - ap^2$ (1)

(ii) EQN OF TANG FROM Q

$Y = qx - aq^2$

$\therefore PX - ap^2 = qx - aq^2$

$(P - q)x = a(p^2 - q^2)$

$x = \frac{a(P - q)(P + q)}{(P - q)}$

$x = a(P + q)$

$\therefore y = qa(P + q) - aq^2$

$y = apq + aq^2 - aq^2$

$y = apq$

$Pt (a(P + q), apq)$

(iii) grad from P, p
grad from Q, q

$\therefore \tan \theta = \frac{P - q}{1 + pq}$

But $\theta = 45$, $\tan \theta = 1$

$1 = \frac{P - q}{1 + pq}$

$1 + pq = P - q$ (1)

(b) $V = \pi \int_{\pi/6}^{\pi/3} (\tan x + \sec x)^2 dx$
 $= \pi \int_{\pi/6}^{\pi/3} \tan^2 x + 2 \tan x \sec x + \sec^2 x dx$
 $= \pi \int_{\pi/6}^{\pi/3} 2 \tan x \sec x + 2 \sec^2 x - [\tan^2 x = \sec^2 x]$

$= \pi [2 \sec x + 2 \tan x - x]$

$= \pi \left[(4 + 2\sqrt{3} - \pi/3) - \left(\frac{4}{\sqrt{3}} + \frac{2}{\sqrt{3}} - \pi/6 \right) \right]$

$= \pi [4 - \pi/6] u^3$ (3)

(c) (i) $T_3 = 10 = a + 2d$ (A)

$T_5 = 4 = a + 4d$ (B)

(B) - (A) $2d = -6$ $d = -3$

$\therefore 10 = a - 6$ $a = 16$

(ii) $S_n = \frac{n}{2} (2a + (n-1)d)$

$0 = \frac{n}{2} (32 + (n-1)(-3))$

$0 = 16n - \frac{3n^2}{2} + \frac{3n}{2}$

Question 4 Cont.

$$-3n^2 - 35n = 0$$

$$n(3n - 35) = 0$$

$$n = 0 \text{ REJECT}$$

$$\text{or } n = 35/3 \therefore n = 12^{th} \text{ term}$$

$$(iii) T_3 = ar^2 = 10 \quad (A)$$

$$T_5 = ar^4 = 4 \quad (B)$$

$$B/A \quad r^2 = 2/5$$

$$r = \pm \sqrt{2/5}$$

$$T_3 = a(\sqrt{2/5})^2 = 10 \quad (2)$$

$$a = 25$$

Question 5.

$$R \sin(x - \alpha) = R(\sin x \cos \alpha - \cos x \sin \alpha)$$

$$R \sin x \cos \alpha - R \cos x \sin \alpha = 2 \sin x - 3 \cos x$$

$$\text{equating } \sin x, \cos x$$

$$R \sin x \cos \alpha = 2 \sin x$$

$$R \cos \alpha = 2 \quad (A)$$

$$R \cos x \sin \alpha = 3 \cos x$$

$$R \sin \alpha = 3 \quad (B)$$

$$\frac{(A)}{(B)} \quad \tan \alpha = 1/5$$

$$\alpha = 56.31^\circ$$

$$\frac{(A)^2 + (B)^2}{R^2} (\sin^2 \alpha + \cos^2 \alpha) = \frac{2^2 + 3^2}{R^2}$$

$$R = \sqrt{13} \quad (2)$$

$$2 \sin x - 3 \cos x = 2.5$$

$$\sqrt{13} \sin(x - 56.31^\circ) = 2.5$$

$$\sin(x - 56.31^\circ) = \frac{2.5}{\sqrt{13}}$$

$$x - 56.31^\circ = 43.8978^\circ, 136.1021^\circ$$

$$x = 100.2078^\circ, 192.4121^\circ \quad (2)$$

$$x = 100^\circ 12', 192^\circ 25'$$

$$(b) (i) n(7) = 6 \text{ (WITH TWO DICE)}$$

$$n(5) = 36 \text{ (36 OUTCOMES)}$$

$$\therefore P(7) = \frac{6}{36} = \frac{1}{6} \quad (1)$$

$$(ii) \left(\frac{7}{6} + \frac{7}{6}\right)^7 = {}^7C_0 \left(\frac{5}{6}\right)^7 + {}^7C_1 \left(\frac{5}{6}\right)^6 \left(\frac{1}{6}\right)^1 + {}^7C_2 \left(\frac{5}{6}\right)^5 \left(\frac{1}{6}\right)^2$$

$$= P(\text{EXACTLY 2 '7's}) = 21 \cdot \frac{5^5}{6^7}$$

$$P(7's) = 0.2344 \quad (2)$$

$$(c) \text{Term } n = {}^7C_{n-1} 2^{8-n} 3^{n-1}$$

$$\text{Term } n+1 = {}^7C_n 2^{7-n} 3^n$$

$$\text{Require } \frac{T_{n+1}}{T_n} \geq 1$$

$$\frac{7!}{(7-n)! n!} 2^{7-n} 3^n \geq 1$$

$$\frac{7!}{(8-n)! (n-1)!} 2^{8-n} 3^{n-1}$$

$$\frac{(8-n) \cdot 3}{n} \geq 1$$

$$24 - 3n \geq 2n$$

$$24 \geq 5n$$

$$n \leq 4.8$$

$$\text{Let } n = 4$$

$$\therefore n+1^{th} \text{ term is the } 5^{th} \text{ term}$$

$$= {}^7C_4 2^3 3^4 \quad (2)$$

$$(d) (i) \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2\cos^2 \theta - 1$$

$$\therefore (ii) \cos 36^\circ = 2(\cos 18^\circ)^2 - 1$$

$$= 2\left(\frac{\sqrt{5}+1}{4}\right)^2 - 1$$

$$= \frac{6+2\sqrt{5}}{8} - 1$$

$$= \frac{6+2\sqrt{5}-8}{8}$$

$$= \frac{2\sqrt{5}-1}{8}$$

$$= \frac{\sqrt{5}-1}{4} \quad (2)$$

Question 6

$$(a) \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}$$

S.1 Prove true for $n=1$

$$\text{LHS} = \frac{1}{2!} = \frac{1}{2}$$

$$\text{RHS} = 1 - \frac{1}{2!} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\text{LHS} = \text{RHS} \text{ TRUE for } n=1$$



S.2. Assume true for $n=k$

$$\frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!}$$

PROVE TRUE FOR $n=k+1$

$$\text{RHS} = 1 - \frac{1}{(k+2)!}$$

$$\text{LHS} = \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= 1 - \frac{k+2}{(k+2)!} + \frac{k+1}{(k+2)!}$$

$$= 1 - \frac{1}{(k+2)!} = \text{RHS} \quad (3)$$

S.3. Step 1 implies true for $n=1$ Step 2 implies if true for $n=1$ true for $n=2, 3, 4, \dots$

$\therefore$ By the Principle of Mathematical Induction true for all $n \geq 1$.

(b) MUST PROVE $\triangle OPQ \equiv \triangle ORQ$

CONNECT O TO P and O TO R.

IN $\triangle OPQ$, $\triangle ORQ$

OP = OR (RADII)

$\angle POQ = \angle ROQ$ (STAND ON EQUAL ARCS)

OQ IS COMMON

$\therefore \triangle OPQ \equiv \triangle ORQ$ (SAS)

$\therefore PQ = RQ$ (MATCHING SIDES IN $\triangle$'s)

$\therefore$ O BISECTS CHORD PR

$\therefore$ OQ $\perp$ PR

IN: QUADRILATERAL OUVW

$$\angle OUV = 90^\circ \text{ (above)}$$

$$\angle VWO = 90^\circ \text{ (data)} \quad (3)$$

$\therefore$ OUVW IS CYCLIC QUAD

OPPOSITE ANGLES SUPPLEMENTARY.

$$(c) (i) f(x) = \frac{2x^2}{x^2+1}$$

$$f(a) = \frac{2a^2}{a^2+1}$$

$$f(-a) = \frac{2(-a)^2}{(-a)^2+1} = \frac{2a^2}{a^2+1}$$

$$f(a) = f(-a) \therefore \text{EVEN} \quad (1)$$

$$(ii) f'(x) = \frac{(x^2+1)(4x) - (2x^2)2x}{(x^2+1)^2} = \frac{4x}{(x^2+1)^2}$$

$$f'(x) \geq 0 \quad x \geq 0 \quad (2)$$

$\therefore$ MONOTONIC INCREASING

$$(iii) \text{ let } y = \frac{2x^2}{x^2+1}$$

$$\therefore \text{ INVERSE } x = \frac{2y^2}{y^2+1}$$

$$y^2x + x = 2y^2$$

$$x = 2y^2 - y^2x$$

$$y^2(2-x) = x$$

$$y = \sqrt{\frac{x}{2-x}} \quad (2)$$

$$(iv) \text{ Domain } 0 \leq x < 2$$

$$\text{Range } y \geq 0 \quad (1)$$

Question 7

$$(a) v^2 = 24 + 8x - 2x^2$$

$$(i) a = \frac{dv^2}{dx} = \frac{d(24+8x-2x^2)}{dx} = 4 - 2x$$

$$a = -2(x-2)$$

$$\therefore \ddot{x} = -\omega^2(x-2)$$

$\therefore$ SIMPLE HARMONIC (2)

$$(ii) x = 2 \quad (1)$$

$$(iii) \text{ FOR MAX AMPLITUDE } v=0$$

$$0 = 24 + 8x - 2x^2$$

$$0 = x^2 - 4x - 12$$

$$0 = (x-6)(x+2)$$

$$x = 6 \quad x = -2$$

$$\text{AMP} = 4. \quad (1)$$

$$(iv) \text{ PERIOD } T = \frac{2\pi}{\omega}$$

$$= \frac{2\pi}{\sqrt{2}}$$

$$= \sqrt{2}\pi \text{ sec} \quad (1)$$

$$(v) \text{ MAX VEL } x=2$$

$$v^2 = 24 + 8(2) - 2(2^2)$$

$$v^2 = 32$$

$$v = 4\sqrt{2} \text{ m/s} \quad (1)$$

$$(b) (i) x = vt \cos \theta \quad 1.$$

$$y = vt \sin \theta - \frac{1}{2}gt^2 \quad 2.$$

$$\text{From 1. } t = \frac{x}{v \cos \theta}$$

$$\text{SUBST 2. } y = \frac{vx \sin \theta}{v \cos \theta} - \frac{g}{2} \frac{x^2}{v^2 \cos^2 \theta}$$

$$y = \tan \theta x - \frac{10}{2} \frac{x^2 \sec^2 \theta}{30^2}$$

$$y = (\tan \theta)x - \frac{\sec^2 \theta}{180} x^2 \quad (2)$$

$$(ii) \text{ when } x=60 \quad y=1.2$$

$$1.2 = 60 \tan \theta - \frac{(\tan^2 \theta + 1) \cdot 3600}{180}$$

$$1.2 = 60 \tan \theta - 20 \tan^2 \theta - 20$$

$$20 \tan^2 \theta - 60 \tan \theta + 21.2 = 0 \quad (2)$$

$$(iii) \tan \theta = \frac{60 \pm \sqrt{3600 - 4(20)(21.2)}}{40}$$

$$\tan \theta = \frac{60 \pm \sqrt{1904}}{40}$$

$$22^\circ 15' < \theta < 68^\circ 54' \quad (1)$$

$$(iv) x_{\max} = \frac{v^2}{g}$$

$$= \frac{30^2}{10}$$

$$= 90 \text{ m}$$

$\therefore$ 30 metres beyond fence.

END OF SOLUTIONS

$$x_{\max} \text{ when } \theta = 45$$

$$R = v^2 \sin 2\theta \quad *$$